

CentaurBench:

Benchmarking LLM Capabilities on Augmenting vs. Automating Real-World Work Tasks



PRESENTED BY

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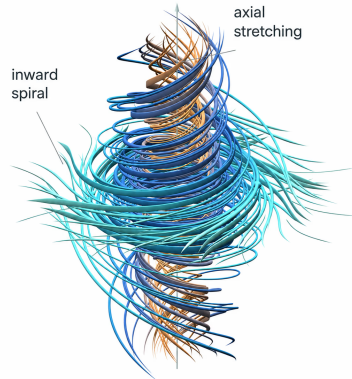
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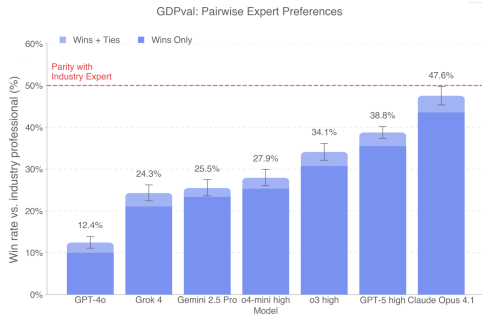
AI is progressing rapidly

OpenAI claims huge maths breakthrough on a famed 'Millennium Problem'

The company says it has cracked the physics of fluids using AI, solving one of the toughest puzzles in mathematics.



AI systems match/exceed experts on real work about half the time

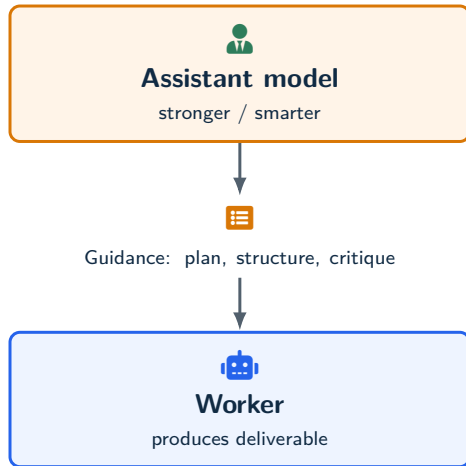


- Top models match or exceed experts on **~50%** of task categories (GDPval 2025)
- Specialist-level performance across medicine, law, coding, and finance (OccuBench 2026, ProfBench 2025, SOPBench 2025)

We want to test models working with other (human or AI) agents

How AI is actually deployed today

- A stronger model helps a weaker model or human complete a task
- The assistant provides planning, structure, or critique
- The worker produces the final output



In this world



**The best player
is not always
the best coach.**

What we need: a research question

Old question

Which model produces
the best output?

Our question

Which model most improves
the work of another agent
on real work tasks?

Our framework

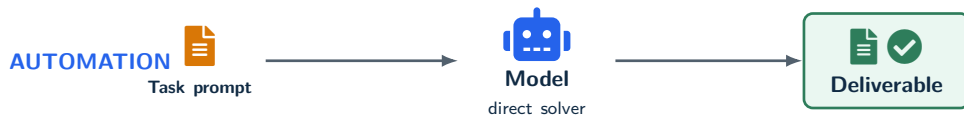
9 frontier models: GPT-5-Mini ▪ GPT-OSS-120B ▪ GPT-O4-Mini ▪ GPT-4.1 ▪ GPT-O3-Mini ▪ Gemini-3.1-Pro ▪ Claude-Opus-4.8 ▪ Claude-Sonnet-4.6 ▪ DeepSeek-V3.1

7 professional tasks: Menu planning ▪ Travel planning ▪ Market trends ▪ Operations research ▪ Tax preparation ▪ Counseling ▪ Tutoring

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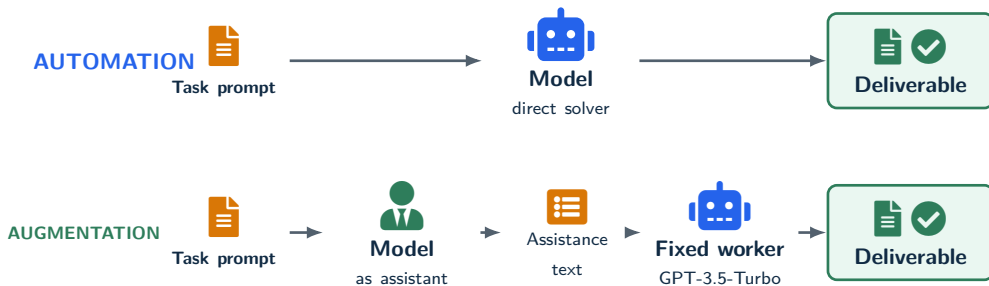
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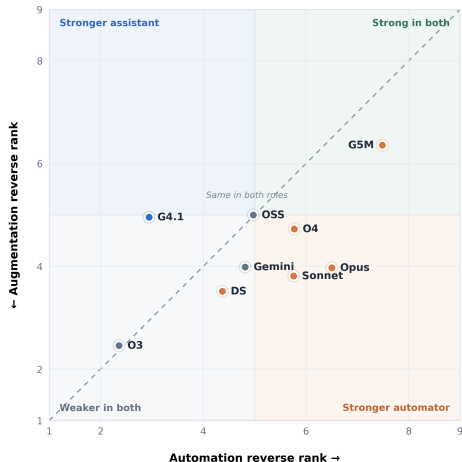
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Preview: the best automator is not the best assistant



Key findings

- The #1 automator loses augmentation on **5 of 7** tasks
- Rank correlation across modes: Spearman $\rho \approx 0.48$ (n.s.)
- The effect is highly **task-specific** (ρ ranges from -0.04 to 0.85)
- We explore qualitatively **why** this occurs

What do we do exactly?

1

Methods

Tasks, experimental
modes, and evaluation

2

Results

By usage mode, and by
task

3

Conclusion

What automation-only
evaluation misses

Tasks

Task	Source	Task type	Deliverable
Menu planning	GDPval	Structured planning	Seven-day constrained meal plan
Travel planning	GDPval	Structured planning	Tokyo itinerary under a fixed budget
Market trends	GDPval	Professional / analytical	Client brief on natural-gas trends
Operations research	Anthropic Economic Index	Professional / analytical	Executive logistics-optimization memo
Tax preparation	Designed task	Professional / analytical	Tax-discrepancy review and correction guidance
Counseling	Anthropic Economic Index	Human-facing / interactive	Supportive counseling-style response
Tutoring	Anthropic Economic Index	Human-facing / interactive	Grade 3 lesson segment

Each task specifies an **observable deliverable**, concrete requirements, and a **task-specific rubric**.

9 focal models × **7 tasks** × **2 modes** × **10 runs**

Running example: Tokyo travel-planning task

Task prompt

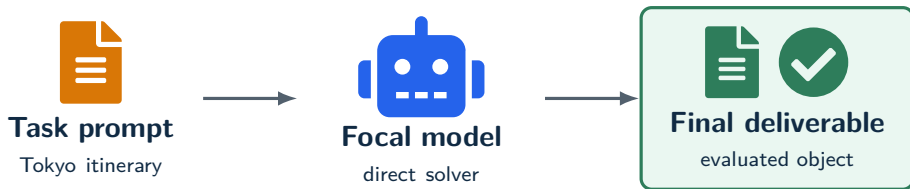
Create a **5-day Tokyo itinerary** for one traveler next month with a total budget of **\$2,500 USD**, excluding passport costs. The traveler is departing from **San Francisco**.

Clearly state assumptions and give clarifying questions **before** the provisional plan. Use realistic estimates, label them as estimates, and keep the plan budget-aware.

We will follow this same task through automation, augmentation, and evaluation.

Experimental modes – Automation

The focal model directly solves the task



Focal models tested: GPT-5-Mini ▪ GPT-OSS-120B ▪ GPT-O4-Mini ▪ GPT-4.1 ▪ GPT-O3-Mini
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Automation final deliverable: Tokyo itinerary

GPT-5-Mini final deliverable (excerpt)

Clarifying questions

- Five full days in Tokyo, or five days including international travel?
- What are the exact dates and accommodation preference?
- Any must-see priorities, dietary restrictions, or mobility constraints?

Assumptions

July 2026; five full days; SFO departure; budget-aware solo traveler; 2–3-star hotel near a major station; 1 USD $\approx$ 155 JPY.

Estimated budget

Flight \$950 Hotel \$600 Transport \$100
Food \$225 Activities \$200 Other \$225

Total: \$2,300 Remaining buffer: \$200

Day-by-day structure

Day 1: Asakusa, Ueno, and Akihabara.

Day 2: Meiji Shrine, Harajuku, Shibuya, and Shinjuku.

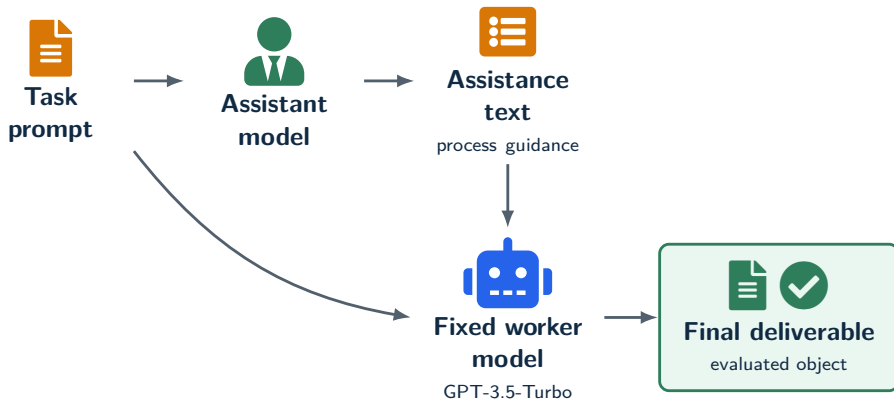
Day 3: Tsukiji, Ginza, and Odaiba.

Day 4: Kamakura day trip, with Hakone as an upgrade.

Day 5: Shopping, flexible sightseeing, and departure.

Experimental modes – Augmentation

The assistant model helps a fixed worker complete the task



Assistant models tested: GPT-5-Mini ▪ GPT-OSS-120B ▪ GPT-O4-Mini ▪ GPT-4.1
GPT-O3-Mini ▪ Gemini-3.1-Pro ▪ Claude-Opus-4.8 ▪ Claude-Sonnet-4.6 ▪ DeepSeek-V3.1

How is the assistance text created?

One universal instruction is used across all seven tasks

Abridged assistance-text creation prompt

*You are not being asked to complete the task. Write **process-only assistance text** for another worker model. The task is provided only so you can help the worker plan its approach.*

Hard constraints

- Do not solve the task or draft the final answer.
- Do not provide task-specific facts, calculations, examples, or recommendations.
- Keep the guidance to roughly 200–250 words.

Required three-phase workflow

1. Requirements Check
2. Plan & Structure
3. Draft → Self-Review → Finalize

Actual assistance text: Tokyo travel planning

GPT-5-Mini assistance text (excerpt)

1. Requirements Check Gather inputs that must be confirmed and list what is missing; compile clarifying questions before drafting; identify hard constraints, deliverables, estimate labels, exclusions, and data needed for realistic costs.

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GPT-5-Mini assistance text (excerpt)

1. Requirements Check Gather inputs that must be confirmed and list what is missing; compile clarifying questions before drafting; identify hard constraints, deliverables, estimate labels, exclusions, and data needed for realistic costs.

2. Plan & Structure Sketch morning/afternoon/evening slots and transport segments; define budget categories and allocation rules; state assumptions up front; create substitution and cost-tradeoff rules.

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3. Draft → Self-Review → Finalize Begin with questions and assumptions, then the day-by-day plan and budget. Check arithmetic, timing, estimate labels, constraints, tone, and clarity; update volatile prices, add contingency, and list outstanding questions.

How do we compare the final deliverables?



Same mode

Automation faces automation;
augmentation faces
augmentation.

Evaluation

How do we compare the final deliverables?



Same mode

Automation faces automation;
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Blind the pair

Randomized A/B order; model
identities are hidden.

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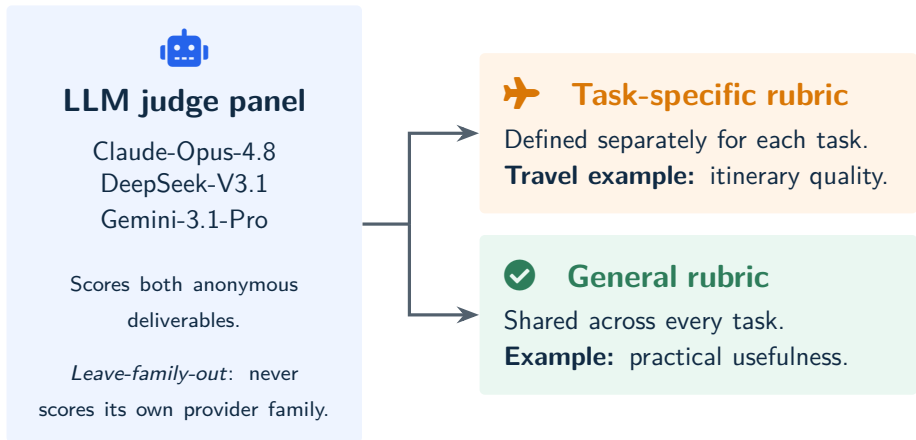


Output evaluation

Pairwise LLM-as-judges with
rubric-based scoring.

Evaluated: final deliverable
Not evaluated: assistance text

How does each LLM judge pick a winner?



LLM as Judge and Ranking Algorithm

NOTATION t : task m : mode r : run y_i : output j : judge $p(x)$: provider family

Fix one task t , mode m , and run r .

1. Collect the ten anonymous outputs:

$$\mathcal{C}_{tmr} = \{y_1, \dots, y_{10}\}.$$

What each operation does

Complete round robin

Every candidate faces every other candidate exactly once per judge when that judge is eligible. No matchup is sampled or omitted.

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$$\mathcal{P} = \{\{i, k\} : 1 \leq i < k \leq 10\}, \quad |\mathcal{P}| = 45.$$

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3. For each pair and judge, sample a random A/B presentation order.

What each operation does

Randomized option order

Draw

$\pi_{ikj} \sim \text{Uniform}\{(i, k), (k, i)\}$. The same pair can therefore appear as either A–B or B–A, reducing position bias.

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3. For each pair and judge, sample a random A/B presentation order.
4. Send the pair only to judges eligible under leave-family-out masking.

What each operation does

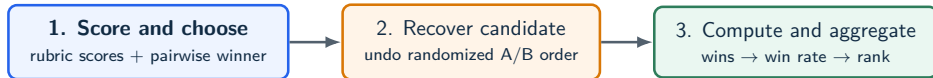
Leave-family-out mask

For candidate pair (i, k) :

$$\mathcal{J}_{ik} = \{j : p(j) \notin \{p(y_i), p(y_k)\}\}.$$

Eligible judges come from neither candidate's provider family.

LLM as Judge and Ranking Algorithm

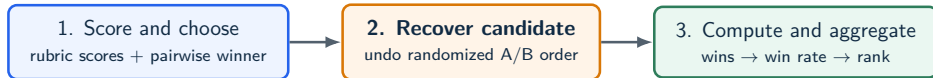


Stage 1: one eligible judge scores one anonymous pair

For judge j and candidate pair $\{y_i, y_k\}$ within fixed task t , mode m , and run r :

- score both deliverables on the task-specific and general rubrics;
- choose the stronger A/B response and provide a short rationale.

LLM as Judge and Ranking Algorithm



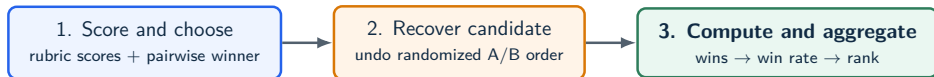
Stage 2: undo the randomized A/B order

After judging, map the chosen A/B response back to its candidate identity, y_i or y_k .

randomized A/B winner $\longrightarrow$ winning candidate y_i or y_k

Count one win for the recovered candidate. Across eligible matchups, these counts become the win-rate numerator in Stage 3.

LLM as Judge and Ranking Algorithm



Within fixed (t, m, r, j)

For candidate y_i ,

$$WR_{i,j}^{(t,m,r)} = \frac{\# \text{ eligible wins by } y_i}{\# \text{ eligible matchups containing } y_i}$$

$WR_{i,j}^{(t,m,r)}$ **Pairwise win rate** for candidate y_i .

$R_{i,j}^{(t,m,r)}$ **Rank induced by that win rate;** rank 1 is best.

Then aggregate the ranks

1. Average $R_{i,j}^{(t,m,r)}$ across eligible judges j .
2. Average across the ten runs r .
3. Combine the seven task rankings into one ordering for each mode m .

Ten-run aggregation and uncertainty

Repeat the full pipeline independently for $r = 1, \dots, 10$.

Panel rank within each run

Let $\mathcal{J}_i$ be the eligible judge set for candidate y_i .

1

$$\tilde{R}_{i,r}^{(t,m)} = \frac{1}{|\mathcal{J}_i|} \sum_{j \in \mathcal{J}_i} R_{i,j}^{(t,m,r)}$$

Average across ten independent runs

2

$$\bar{R}_i^{(t,m)} = \frac{1}{10} \sum_{r=1}^{10} \tilde{R}_{i,r}^{(t,m)} \quad \text{SE}_i^{(t,m)} = \frac{\text{SD}_r(\tilde{R}_{i,r}^{(t,m)})}{\sqrt{10}}$$

3

Compute the rank of ranks

Order models by their unrounded ten-run mean ranks $\bar{R}_i^{(t,m)}$.

Pairwise Judge and Rubric Scoring Example

Travel-planning scorecard

Augmentation, Run 10

A: Claude-Sonnet-4.6 **B: GPT-5-Mini**

TASK-SPECIFIC

Completeness	5.0	5.3
Cost realism and arithmetic	3.5	4.4
Itinerary quality	6.2	4.2

GENERAL

Instruction following	5.8	6.3
Accuracy and specificity	5.2	3.7
Practical usefulness	5.2	4.2

6 of 11 rubric dimensions shown for brevity.

Gemini-3.1-Pro chose Claude-Sonnet-4.6

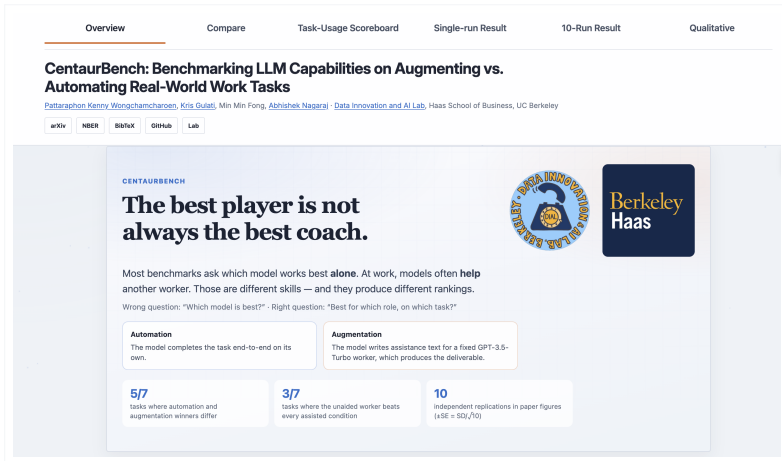
More geographically sensible and realistic; avoids closed attractions and leaves a plausible share of the \$2,500 budget for flights.

DeepSeek-V3.1 chose GPT-5-Mini

More complete, handles uncertainty with clarifying questions, and follows the instructions more closely, despite weak flight-cost accounting.

Choices determine win rates. Scores and rationales explain why.

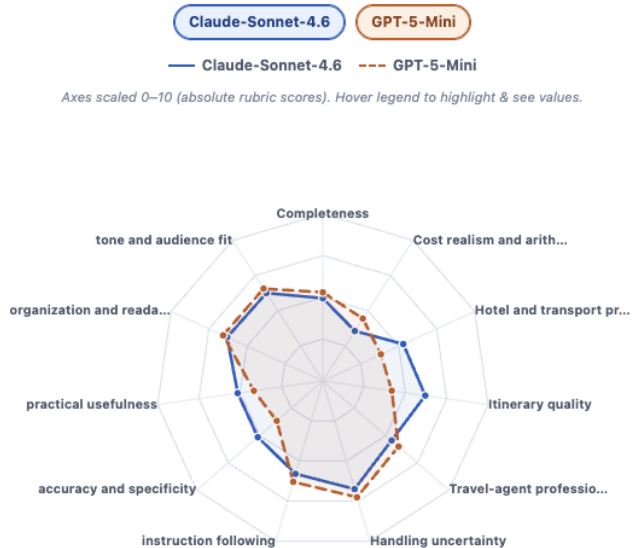
CentaurBench dashboard



kennywong524.github.io/centaur-benchmark/

Compare models, modes, tasks, runs, deliverables, rubric scores, and judge rationales.

Example dashboard comparison



Three views of model performance

1

Results by mode

Automation versus
augmentation rankings

2

Results by task

Rank correlation varies across
tasks

3

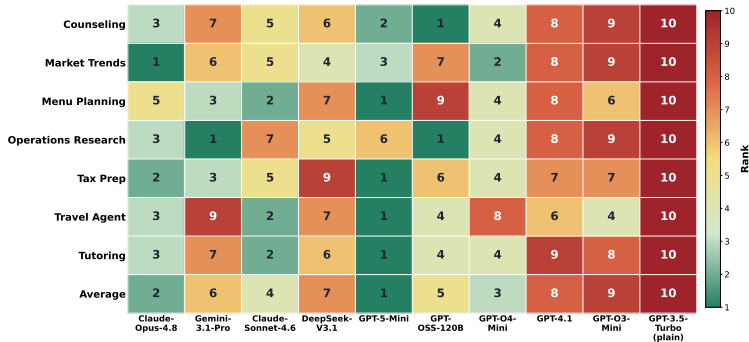
Qualitative analysis

Why assistance succeeds or
fails

Results by mode

Average row:
$$\bar{R}_i^{(\text{avg}, m)} = \frac{1}{10} \sum_{r=1}^{10} \left(\frac{1}{7} \sum_{t=1}^7 \tilde{R}_{i,r}^{(t,m)} \right) = \frac{1}{70} \sum_{r=1}^{10} \sum_{t=1}^7 \tilde{R}_{i,r}^{(t,m)}.$$

Automation

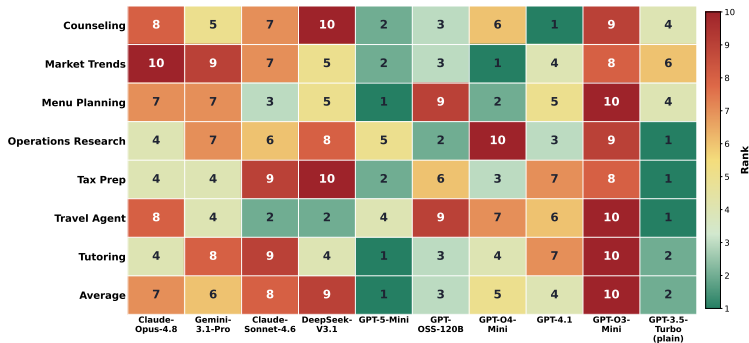


GPT-5-Mini ranks first on average and wins four tasks.

Results by mode

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Augmentation



No assistant dominates; the unaided worker wins three tasks.

Results by task

Task	Best automator	Best augmentation condition
Counseling	GPT-OSS-120B	GPT-4.1
Market trends	Claude-Opus-4.8	GPT-O4-Mini
Menu planning	GPT-5-Mini	GPT-5-Mini
Operations research	Gemini-3.1-Pro / GPT-OSS-120B	Unaided worker
Tax preparation	GPT-5-Mini	Unaided worker
Travel planning	GPT-5-Mini	Unaided worker
Tutoring	GPT-5-Mini	GPT-5-Mini

Results by task

$$\rho_t = \text{corr}_{\text{Spearman}}\left(\overline{\mathbf{R}}^{(t,\text{auto})}, \overline{\mathbf{R}}^{(t,\text{aug})}\right)$$

$\overline{\mathbf{R}}^{(t,m)} = \left(\overline{R}_i^{(t,m)}\right)_{i=1}^9$ is the ten-run mean-rank vector across the nine shared models.

Task	Spearman ρ_t
Tax preparation	0.85
Menu planning	0.64
Tutoring	0.50
Counseling	0.25
Operations research	0.17
Market trends	0.10
Travel planning	-0.04

0.85

Automation and augmentation rankings have a strong positive rank correlation in tax preparation.

Results by task

$$\rho_t = \text{corr}_{\text{Spearman}}\left(\overline{\mathbf{R}}^{(t,\text{auto})}, \overline{\mathbf{R}}^{(t,\text{aug})}\right)$$

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-0.04

Automation and augmentation rankings are essentially unrelated in travel planning.

Qualitative analysis

Why do some assistance texts help while others hurt?

Specificity beyond the prompt

Adds analytical steps beyond what the prompt specifies

Explicit structural requirements

Prescribes section sequence and output format explicitly

Direction maintenance

Keeps the worker oriented toward task completion

Effective assistance: specific, structured, and directional
Ineffective assistance: generic, unstructured, or misdirecting

Next steps

What can the
model do?



What can the
model help?



Validate with people

Human and domain-expert
validation



Study richer assistance

Multi-turn support; more
workers

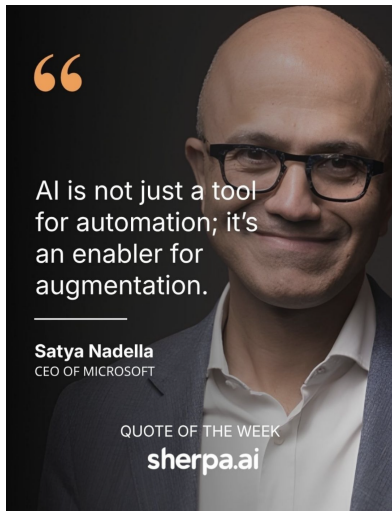


Broaden real-world work

More tasks and work settings

**We need to align measurement with what we actually want to do
with AI!**

If we want augmentation, we must measure it!



“

**If you can't measure it,
you can't improve it.**

Peter Drucker

Thank you



Data Innovation & AI Lab



Dashboard

Questions?

DIAL website

thedatainnovationlab.com

Email

data-innovation-lab@berkeley.edu

Feedback and collaborators are welcome!